

Identified Needs

This section presents the needs we identified from the twelve interviews. Participant counts are based on the interview summaries, not on a formal coding of transcripts. The participant key at the end maps the codes P1 to P12 to interviews.

Key needs

Need 1: Short, step-by-step help that goes deeper on request

AI explanations are often too long or too in-depth and arrive all at once. Students want a short first answer, a check that the question was understood, and then the option to go deeper.

- **Evidence (4 of 12):** P5, P6, P11, P12.

“It’s helpful when the help comes step by step and not all at once.”

— P11

- P12 wants a summary first and, once the question is confirmed, a deeper prompt, because “often it’s so much text and you only need a small part of the explanation”.
- P5 called answers that are too long or complex unhelpful, and P6 wanted more concise support.
- **Why it matters:** a long answer that goes in a slightly different direction than expected confuses the student instead of helping. Several participants said an answer only became useful after multiple prompts, so a tool could be built around that pattern from the start.

Need 2: Help that matches what the student already knows

Help is unhelpful when it assumes knowledge the student does not have yet. The student then has to explain what they know and prompt again.

- **Evidence (4 of 12):** P2, P5, P6, P11.
- P5 and P6 both said the first answer assumes too much prior knowledge. P2 sometimes “lacked the basics” to understand the help and blamed themselves, not the tool. P11 found an explanation too difficult and had to give the AI the model solution to explain instead.
- **Why it matters:** uploading all lecture slides does not solve this. A professor may post 100 slides while the student only needs help with 20 of them, so a tool that assumes all of them are known skips the basics the student is missing. A few quick opening questions about what is understood, and simple examples that build up from basics, could address this.

Need 3: Help grounded in the course material

Students want the AI to know their lecture material and to point back to it instead of bringing in outside content.

- **Evidence (4 of 12):** P3, P4, P9, P11.

“It brought in topics that weren’t from the class to answer a problem.”

— P9

- P3 asked for more subject-specific context (slides, scripts) and for AI to advise where in the course material to look. P4 noted that AI uses correct theorems but formulates them

differently from the script. P11 always uploads lecture notes, because chemistry answers are weaker without them.

- P9 got the most out of a setup where the AI had all of the professor’s notes and pointed to which lecture to look up without giving the solution.
- **Why it matters:** an answer that also points to the slides gives the student two ways to check it, by reading the AI response and by finding the content in professor-approved material. It also keeps answers inside what the exam covers.

Need 4: Support that keeps the student doing the thinking

Students want help that guides and supervises their work instead of solving it for them. A finished answer feels fast but leaves less understanding behind. The strongest pattern is that the student starts the work and the AI follows their reasoning, instead of imposing its own approach.

- **Evidence (10 of 12):** P2, P3, P4, P6, P7, P8, P9, P10, P11, P12.
- P3 wants AI to point to where to look in the course material instead of giving the solution, and P4 finds AI too focused on solutions. P7 said a complete code fix created an illusion of competence that backfired in pen-and-paper exams. P8 adopted an AI explanation without forming their own arguments and could not defend it. P6 named cognitive offloading as a concern.
- P2, P8 and P9 already use AI to check their own steps. P10 prefers being walked through the line of thinking, and P11 feels they solve the problem themselves when the AI gives a formula and they do the calculation.

“I was still trying to solve the problem, even with help.”

— P9

- **Why it matters:** when students only ask for the next part, they cannot tell whether their own work is correct, and the AI never sees their method. If the AI first walks through the student’s attempt, it can correct them along the way, which is especially useful for open-ended questions. A formula can be fine to receive, but students still need to understand where it comes from.

Need 5: Knowing when not to trust the AI

Students need support judging when an AI answer may be wrong. Trust varies strongly between participants, and both extremes cause problems.

- **Evidence (5 of 12):** P1, P2, P4, P11, P12.

“My peers make mistakes, unlike AI, hence I go directly to it.”

— P2

- P1 trusts AI completely, over friends, and could not submit homework after AI-generated code failed and they could not fix it. P11 reported hallucinated answers and wrong source references, and uses another tool for sources. P4 saw AI repeatedly give wrong base cases in a dynamic programming exercise. P12 got poor answers from two assistants on a circuits question and skipped it after 30 minutes.
- **Why it matters:** an AI can be wrong with the same confidence as when it is right, and hallucinations matter more in some fields, such as chemistry or proofs. Pointing to course

material (Need 3) is one way to check an answer, and tools should make checking easier, for example through references.

Secondary needs

- **Pushback before answering.** P1 tries for under a minute before asking for help, and P9 said “I feel like sometimes maybe I give up a bit before I really need to”. P6 realized they could have solved a problem themselves, and P11 wished they had spent more time on problems they wanted to learn. Time pressure pushes students to ask earlier (P5, P7, P8).
- **AI-generated code is hard to understand.** P1 did not know how to correct the code AI gave them.
- **A safe place to ask basic questions.** P9 said “you can ask it the dumbest question and it will answer you”, and P12 is afraid to admit not fully understanding a professor’s answer.
- **Help with formulating the question.** P12 often finds the solution while writing the prompt, and P3 puts effort into phrasing questions. Clarifying what is unclear before studying may be part of the support.
- **The aha moment.** P12 recognizes a problem the next time after having an aha moment, so emphasizing the core idea could help.
- **Breaks.** P11 takes a break or switches problems and comes back with a fresh mind. This is supported by one participant only.
- **Subject dependence.** In STEM, going back and forth with AI resembles working with an expert, and TAs and professors are rarely available. Satisfaction likely depends on the person and subject, and we interviewed only STEM students.
- **Vague or answer-giving help.** P2 said slides, other resources and people are either too vague or directly give the answer, which is why they go to AI first.

Appendix: Participant key

The interview files use interviewer-based codes (for example `franco_1`). This table maps them to P1 to P12. Participants are identified only by role.

Code	Interview code	Interviewed by	Role
P1	<code>franco_1</code>	Franco	Medicine BSc student
P2	<code>franco_2</code>	Franco	Mechanical Engineering graduate
P3	<code>valentin_1</code>	Valentin	Mathematics BSc student
P4	<code>valentin_2</code>	Valentin	Quantitative Finance MSc student
P5	<code>noah_1</code>	Noah	Mechanical Engineering student
P6	<code>noah_2</code>	Noah	Robotics student
P7	<code>rafael_1</code>	Rafael	Computer Science BSc student
P8	<code>rafael_2</code>	Rafael	Information Systems BSc student
P9	<code>nicolas_1</code>	Nicolas	Industrial Engineering student
P10	<code>nicolas_2</code>	Nicolas	Computational Biology student
P11	<code>reto_1</code>	Reto	Pharmacy BSc student
P12	<code>reto_2</code>	Reto	Electrical Engineering student